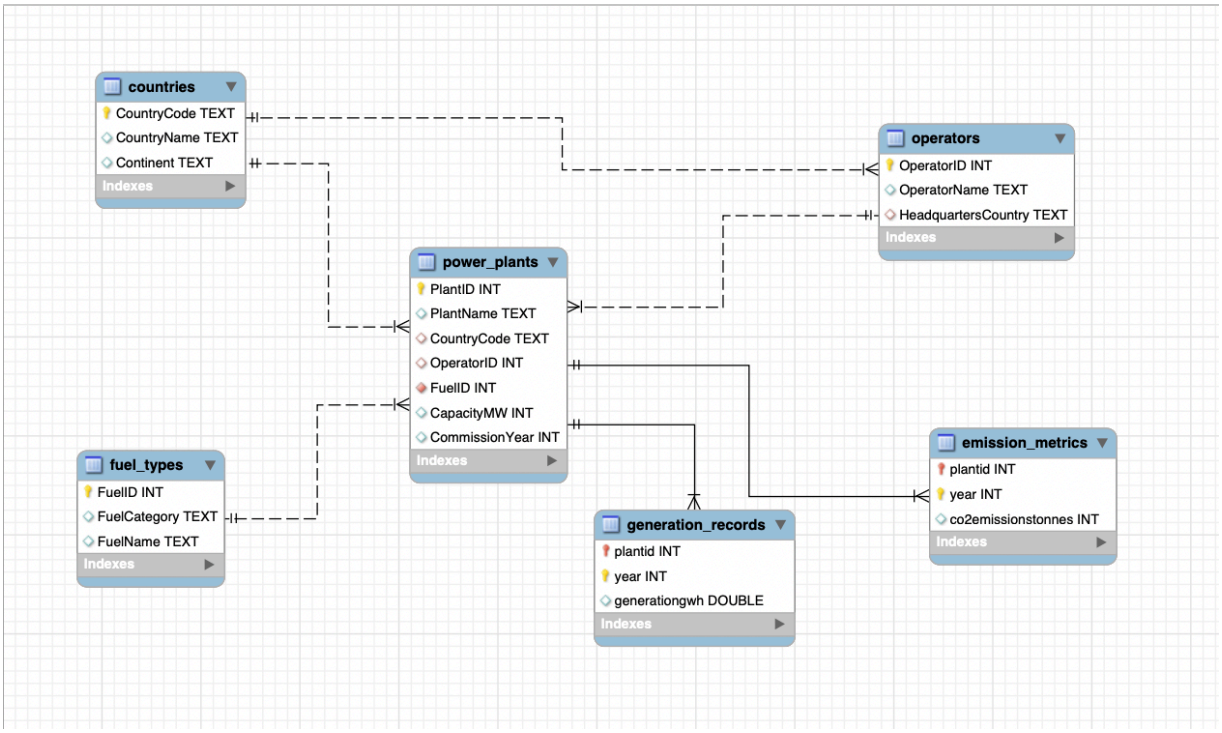
CSC 351- Assignment 3
Madeline Ashton

EER Diagram:



1.

```
19    -- 1. Retrieve the plant name, country name, operator name, fuel category, fuel name,
20    -- capacity (in MW), and commission year for all power plants without using table aliases.
21    -- Sort the results in descending order by capacity.
22 •  select power_plants.PlantName, countries.CountryName, operators.OperatorName,
23    fuel_types.FuelCategory, fuel_types.FuelName, power_plants.CapacityMW,
24    power_plants.CommissionYear
25    from power_plants natural join countries
26                  natural join operators
27                  natural join fuel_types
28    order by power_plants.CapacityMW desc;
29
30    -- 2. Retrieve the plant name, country code, calendar year, and annual generation (in GWh)
```

100% 39:28

Result Grid Filter Rows: Search Export:

PlantName	CountryName	OperatorName	FuelCategory	FuelName	CapacityMW	CommissionYear
Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000	1981
Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

2.

```
30  -- 2. Retrieve the plant name, country code, calendar year, and annual generation (in GWh)
31  -- for all power plants for the year 2024 without using table aliases. Sort the results in
32  -- descending order by generation.
33  select power_plants.PlantName, power_plants.CountryCode,
34  generation_records.year, generation_records.generationgwh
35  from power_plants natural join generation_records
36  where generation_records.year = 2024
37  order by generation_records.generationgwh desc;
38
39  -- 3. Retrieve the plant name, country code, calendar year, annual generation (in GWh),
40  -- and CO2 emissions (in tonnes) for all power plants for the year 2024 without using table al
```

100% 48:37

Result Grid Filter Rows: Search Export:

PlantName	CountryCode	year	generationg...
Three Gorges	CHN	2024	98800
Itaipu	BRA	2024	89000.1
Tuoketuo	CHN	2024	71000.3
Vindhyachal	IND	2024	35600.8
Grand Coulee	USA	2024	21500.5
Janschwalde	DEU	2024	18400.2
Loy Yang	AUS	2024	14200.4
Kashiwazaki-Kariwa	JPN	2024	0

3.

```
    -- 3. Retrieve the plant name, country code, calendar year, annual generation (in GWh),
40  -- and CO2 emissions (in tonnes) for all power plants for the year 2024 without using table aliases.
41  -- Sort the results in ascending order by CO2 emissions.
42  select power_plants.PlantName, power_plants.CountryCode,
43  generation_records.year, generation_records.generationgwh,
44  emission_metrics.co2emissionstonnes
45  from power_plants natural join generation_records
46                  natural join emission_metrics
47  where generation_records.year = 2024
48  order by emission_metrics.co2emissionstonnes asc;
49
50  -- 4. Write a SQL query using a Common Table Expression (CTE) to calculate the total cumulative power
```

100% 1:49

Result Grid Filter Rows: Search Export:

PlantName	CountryCode	year	generationg...	co2emissionstonn...
Grand Coulee	USA	2024	21500.5	0
Itaipu	BRA	2024	89000.1	0
Kashiwazaki-Kariwa	JPN	2024	0	0
Three Gorges	CHN	2024	98800	0
Loy Yang	AUS	2024	14200.4	1950000
Janschwalde	DEU	2024	18400.2	2450000
Vindhyachal	IND	2024	35600.8	3100000
Tuoketuo	CHN	2024	71000.3	5200000

4.

```
-- 4. Write a SQL query using a Common Table Expression (CTE) to calculate the total cumulative power
-- generation (in GWh) for each operator across all available years without using table aliases.
-- Retrieve the operator name, headquarters country, and their total generated power, and sort the results
-- in descending order by total generation.
with cumul_power_gen as(
    select power_plants.OperatorID, sum(generationgwh) as cpg
    from power_plants natural join generation_records
    group by power_plants.OperatorID)

select operators.OperatorName, operators.HeadquartersCountry,
cumul_power_gen.cpg
from operators natural join cumul_power_gen
order by cumul_power_gen.cpg desc;
```

100% 1:63

Result Grid Filter Rows: Search Export:

OperatorName	HeadquartersCoun...	cpg
China Yangtze Power	CHN	169800.3
Electrobras	BRA	89000.1
NTPC Limited	IND	35600.8
Global Hydro Inc	USA	21500.5
Rhein Power AG	DEU	18400.2
Snowy Hydro	AUS	14200.4
Tepco Energy	JPN	0

5.

```
68  with country_total_power as(
69      select sum(generation_records.generationgwh) as country_tpg,
70      power_plants.CountryCode
71      from generation_records natural join power_plants
72      group by power_plants.CountryCode),
73
74  country_total_emission as(
75      select sum(emission_metrics.co2emissionstonnes) as country_tce,
76      power_plants.CountryCode
77      from emission_metrics natural join power_plants
78      group by power_plants.CountryCode)
79
80  select countries.CountryName, country_total_power.country_tpg,
81  country_total_emission.country_tce
82  from countries natural join country_total_power
83              natural join country_total_emission
84  order by country_total_power.country_tpg desc;
```

100% 47:84

Result Grid Filter Rows: Search Export:

CountryName	country_tpg	country_tce
China	169800.3	5200000
Brazil	89000.1	0
India	35600.8	3100000
United States	21500.5	0
Germany	18400.2	2450000
Australia	14200.4	1950000
Japan	0	0